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Fracture Identification, Prediction and Application in Lower Cretaceous Yamama Carbonate Reservoir in Middle East

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Yamama formation is widely distributed in the southeast of Iraq, and many large oilfields have proved that its reserves exceed 25 billion barrels (Figure 1). Yamama reservoir is generally characterized by deep burial (3500-4500 m), high production (up to 3,500 barrels per day), low viscosity (less than 0.5 cp) and abnormal high pressure (pressure coefficient changes from 1.2 to 1.9). Yamama reservoir has good development potential. Few exploration and production wells drilled to Yamama reservoir, scattered seismic data with moderate to poor quality, unclear geological characteristics made it difficult to appraise the reservoir. Preliminary exploration and appraisal results show that most Yamama reservoirs are pore-type, and there is little research on Yamama fractured reservoirs which have high production potential. Through limited data analysis, this study aims to discuss the identification and characterization of Yamama reservoir fractures. It will provide guidance for efficient exploration and development of this reservoir in the future.

Introduction

Yamama formation is widely distributed in the southeast of Iraq, and many large oilfields have proved that its reserves exceed 25 billion barrels (Figure 1). Yamama reservoir is generally characterized by deep burial (3500-4500 m), high production (up to 3,500 barrels per day), low viscosity (less than 0.5 cp) and abnormal high pressure (pressure coefficient changes from 1.2 to 1.9). Yamama reservoir has good development potential. Few exploration and production wells drilled to Yamama reservoir, scattered seismic data with moderate to poor quality, unclear geological characteristics made it difficult to appraise the reservoir. Preliminary exploration and appraisal results show that most Yamama reservoirs are pore-type, and there is little research on Yamama fractured reservoirs which have high production potential. Through limited data analysis, this study aims to discuss the identification and characterization of Yamama reservoir fractures. It will provide guidance for efficient exploration and development of this reservoir in the future.

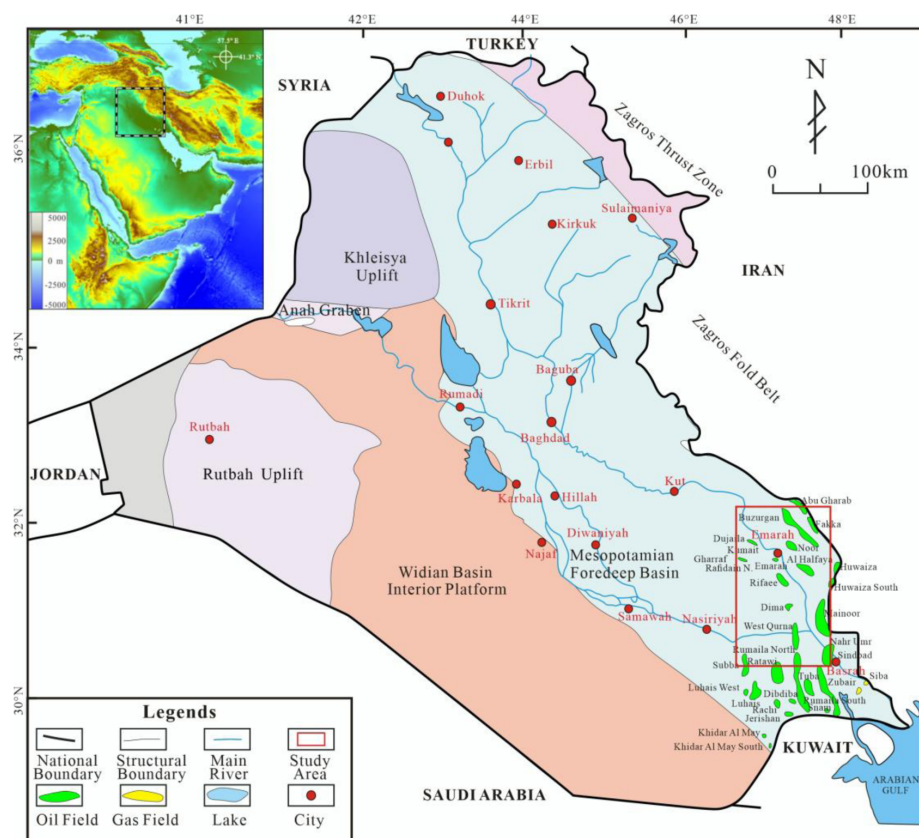


Figure 1—The location of the study area in southeast Iraq

Geological Setting

The study area is located in the southeast of Iraq, which has many large oilfields in the world, such as Rumaila oilfield and West Qurnar oilfield. It belongs to the front belt of Mesopotamian foreland basin related to Zagros orogenic belt. Yamama deepens gradually from southwest to northeast with several anticlines developed. The main oilfields are located in the anticline area. The time span of the lower Cretaceous is about 145 Ma to 113 Ma, belonging to Berriasian stage to Albion stage. The main strata from bottom to top are Sulaiy, Yamama, Ratawi, Zubair and Shuaiba (Figure 2).

There are 11 tectonostratigraphic mega sequences separated by major unconformities and 63 maximum flooding surfaces (MFSs) in Arabia plate, within those eleven mega sequences recognized from biostratigraphic data integrated with sedimentological and lithological data (Sharland, 2001), the author also mentioned that of the 63 MFSs, 33 are interpreted as plate wide, 25 are sub-regionally wide over large area of the plate and 5 are only locally preserved. Yamama is located in the bottom of AP8 sequence (super sequence of Ariba Plate) and can be divided into two progradation- retrogradation third order sequence. The barrier units YB-1 and YB-2 are correlated with MFSs of K20 and K30 in Ariba Plate sequence, they are identified in wells and seismic which separated three reservoir units: YR-A, YR-B and YR-C.

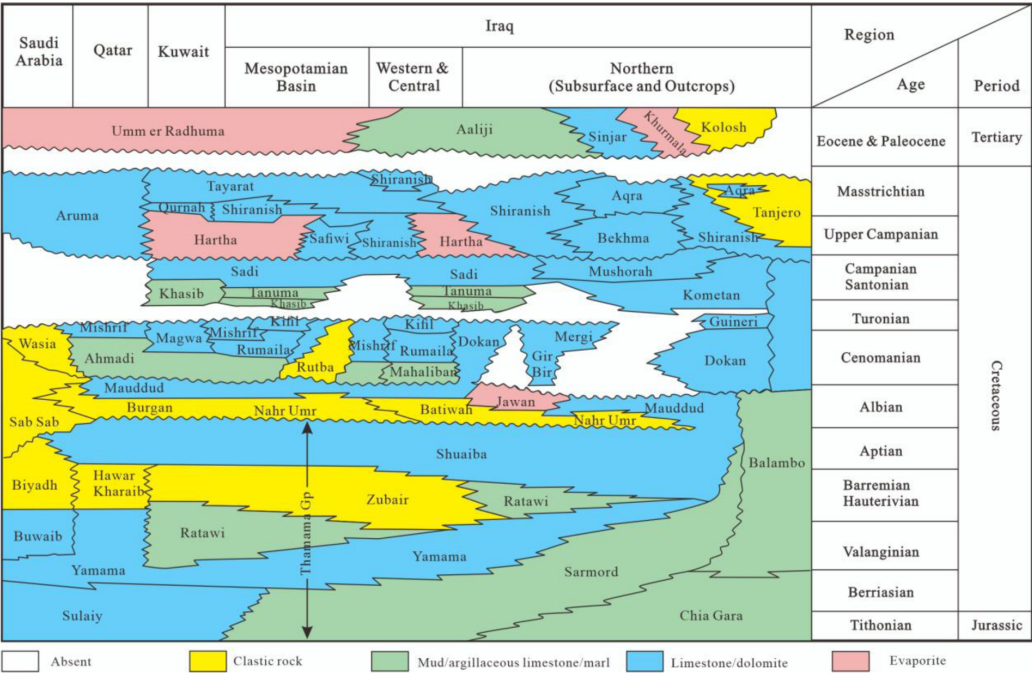


Figure 2—The strata distribution of the Arabian plate

Yamama Formation is widely distributed in the whole southeast of Iraq, but its lithology changes from southwest to northeast. Mud limestone, bioclastic limestone and oolite can be seen in the southwest, but high-energy bioclastic limestone and argillaceous limestone are rare in the northeast, mainly micrite limestone. In southeastern Iraq, the upper part of the Yamama cycle consists of inner carbonate ramp deposits, including bioclastic and oolitic limestones, while the lower part features middle-carbonate ramp deposits dominated by bioclastic and pelagic limestones, and the base part consists of outer carbonate ramp deposits of micrite. Carbonate ramp is the dominated sedimentary environment of Yamama formation and it can be divided into inner ramp, middle ramp and outer ramp. From southeast to northwest (the direction from Ratawi oilfield to Halfaya oilfield), the sediment environment changed from inner ramp to outer ramp with the basin ward progradation. From bottom to top, YR-C dominated by middle or outer carbonate ramp with mudstone and mound developed, YR-B dominated by inner or middle ramp which indicated the progradation of the ramp, YR-A dominated by middle ramp with wackestone developed.

Data and Method

The research data include high resolution seismic data and well data. To better study the characteristics of fractures and effectively predict the distribution of fractures, high resolution seismic data, core and logging data are used in this study. The post-stack 3D seismic data covers H Oilfield, with track pitch 25m*25m, and the main frequency is about 30 Hz. Two cored wells with about 25 m core intervals which covered the main reservoir intervals. The core interval has systematic analysis result. In addition to conventional logging data, there are many special logging data, especially FMI data, which is more conducive to identify fracture.

This study is constrained by regional geological understanding, and the fracture identification by coring wells is the hard data to clarify the fracture development intervals and characteristics. The conventional logging is calibrated by the results of coring well fracture identification, and the fracture development characteristics of special logging interpretation are combined to predict the fracture development by neural network method. 3D seismic was used to identify fractures by multi-method and multi-attribute. Calibrating the development of seismic prediction fractures with the interpretation results of wells, and clarify the most sensitive seismic prediction parameters, and then achieve the 3D prediction result of fracture.

Results

Fracture identification in core wells

Lithological characteristics. Based on the core description, master log and ECS logging results, the main lithology of Yamama are limestone and marl, with few shale. The master log shows thick limestone developed, thin marl interbedded in the thick limestone. Few shales only developed in the lower part of Yamama. The YRA zone is dominated by limestone, mainly consisting of biosparite, biosparitic mudstone, and mudstone. The YRB zone is dominated by limestone and mudstone, while the YRC zone mainly consists of mudstone with some thin layers of limestone. The property of Yamama is tight, based on the core plug data analysis, the porosity range is 1% to 9%, the main porosity is about 3.4%. The permeability range is 0.01 mD to 14.6 mD, the main permeability is about 1.3 mD. This tight reservoir is normally tend to develop fracture. The presence of microfractures and small scale fractures can improve the reservoir petrophysical properties.

Fracture characteristics in core wells. Core description is the direct method to identify fracture. Through detailed observation and description of cores of two cored wells and related analysis results, it shows that: fracture are well developed in Yamama reservoir, but the scale is not too large, and most fracture do not cut through the whole core (Figure 3); From the core photos, it can be seen that the fractures are mostly medium-low angle, especially the low-angle or near horizontal fractures are developed. Micro-fractures can be seen in typical thin sections, most of the micro-fractures are open with small scale. The fracture density changes from 2 to 12 per meter, the average fracture density is about 5 per meter. The fracture shows obvious hydrocarbon characteristics under ultraviolet light. Most fractures showed oil cut, which is good channel for oil migration and accumulation.

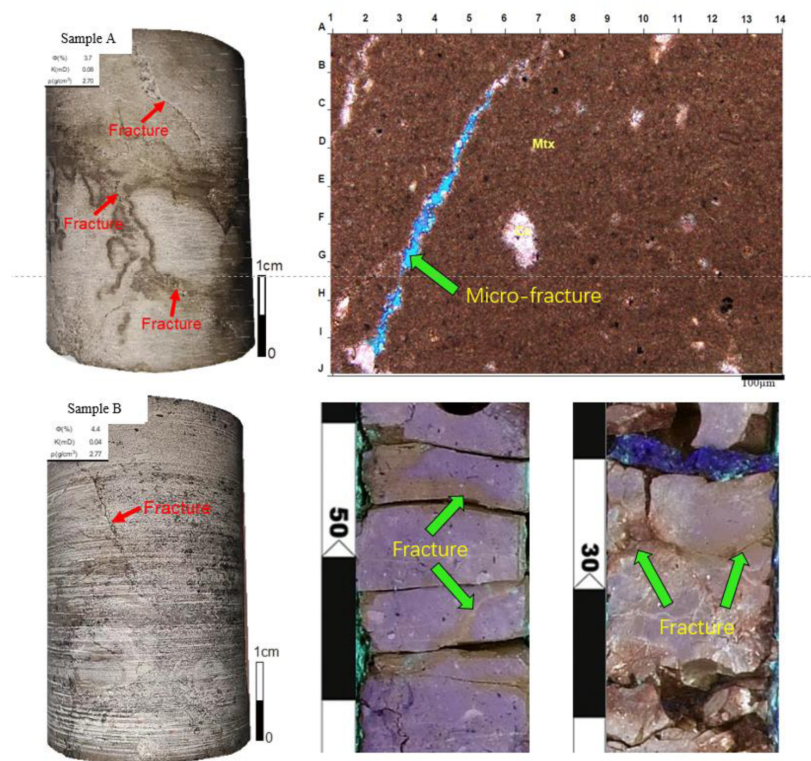


Figure 3—Fracture identification result of core in Yamama reservoir

Fracture identification and distribution in wells

Fracture identification in special logging. Special logging, especially FMI logging, is an effective method to identify and characterize fractures in wells. Three wells have FMI data in the study area, which will be beneficial to the research on fracture identification. All wells showed fracture developed in Yamama formation. For example, the FMI interpretation results of typical well A shows that fracture zones are developed in 4343m-4345m intervals, the angle of the fracture is about 10-20 degrees. Through the comparison of coring intervals, core breakages and fractures can be seen in these intervals (Figure 4).

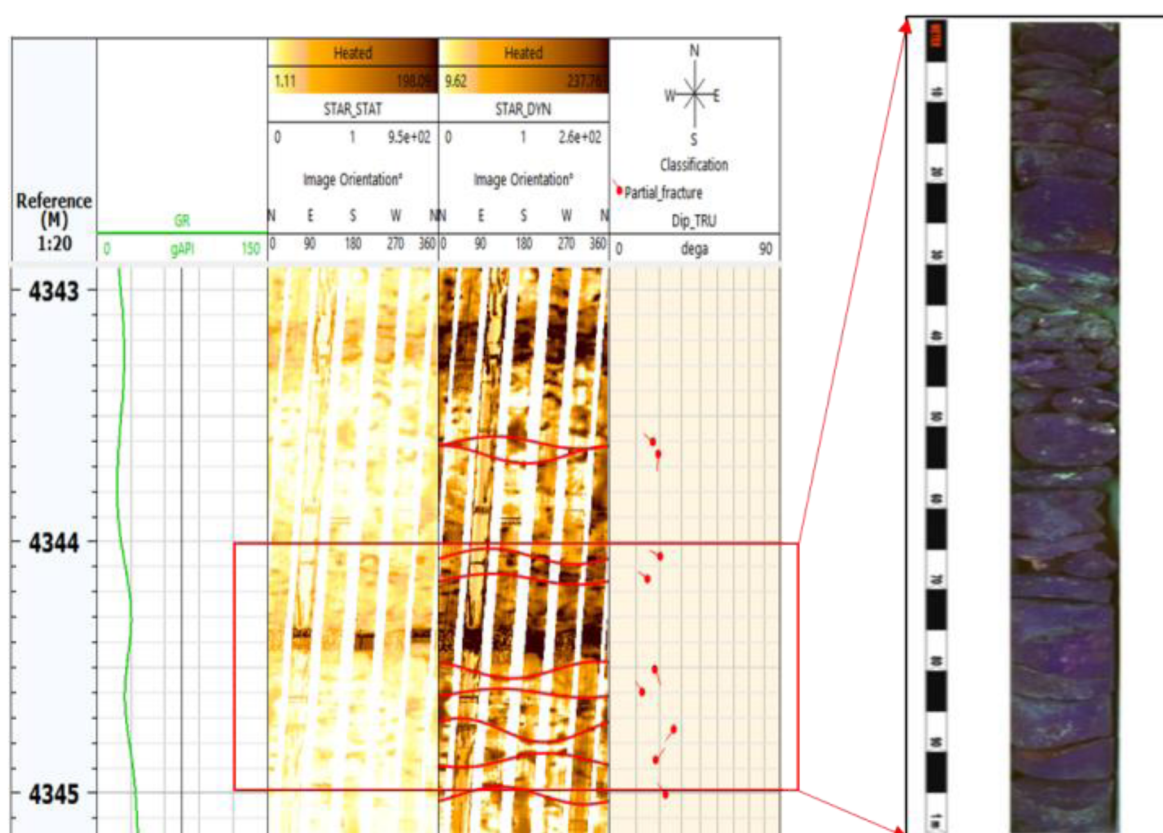


Figure 4—Fracture identification result of FMI logging in well A

Through the further analysis of FMI logging data, the parameters of the fracture can be obtained (Table 1). According to the FMI interpretation results, the degree of fracture development in each well has big difference. Well B has the highest degree of fracture development. There are 40 pieces of fractures can be identified in this well, the other wells have less than 10 pieces. The average aperture change from 0.0202 mm to 0.0214 mm with minor difference. While the average density and average length have big change, the average density change from 1.3/m to 4.0/m, the average length change from 1.49/m to 4.29/m. Well A has the highest fracture density and length. The calculated average fracture porosity changes from 0.0024% to 0.0077%, and the overall fracture porosity is small. The comprehensive interpretation based on FMI special logging can also get the dip distribution characteristics of fractures, most of which are between 10-40 degrees, and the whole fracture is low angle.

Table 1—Fracture parameters interpretation result of FMI

Well	Number (Piece)	Ave. Aperture (mm)	Ave. Density (1/m)	Ave. Length (1/m)	Ave. Porosity (%)
A	7	0.0213	4.00	4.29	0.0077
B	40	0.0214	2.66	2.74	0.0061

Well	Number (Piece)	Ave. Aperture (mm)	Ave. Density (1/m)	Ave. Length (1/m)	Ave. Porosity (%)
C	5	0.0203	1.30	1.49	0.0024

The response characteristics of fractures can also be seen in conventional logging. By comparing the core analysis results, we can see that core with fracture permeability (red line) are significantly greater than core without fracture permeability (black line) when same porosity (Figure 5). The big fractures have not developed in Yamama due to non-mud loss during the drilling from masterlog. It is difficult to identify the middle-low angle fracture, especially the near horizontal fracture in Yamama reservoir. Based on the calibration of logging response characteristics and core description result, the invasion may occur during and after the drill which will obviously show the difference between LLD and LLS logging. The principle of this method is mainly based on the degree of mud invasion to identify the possible fracture development. Generally, mud invasion in tight reservoirs is small. When fractures develop, it will cause mud invasion to a certain extent, which will lead to the different response between LLD and LLS. The possible fracture development position can be identified by the difference between LLD and LLS.

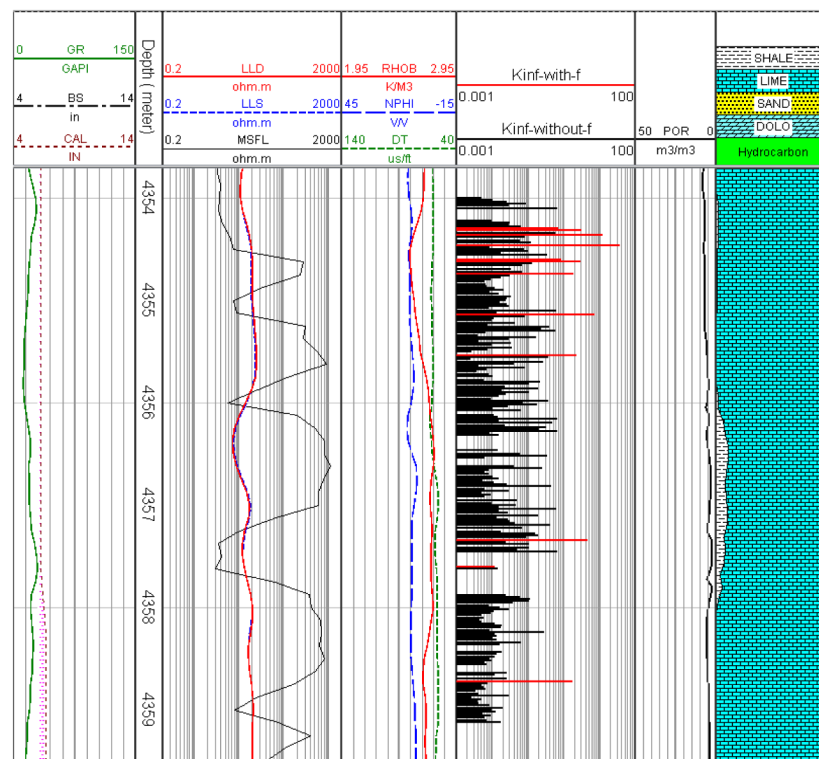


Figure 5—Permeability of samples with fracture and without fracture in typical core well

Based on this logging response characteristics, the fracture development characteristics are achieved through the comprehensive interpretation results of multiple wells (Figure 6): the fracture are mainly developed between Well B and Well E, which are located in the structure core and wing area, and the fractures out of the structure core area are not well developed; Vertically, the fracture development interval is located in the middle and upper of Yamama formation; The fracture distributed in a large area, mostly between 2-3 well spacing, indicating that the fracture has good connectivity.

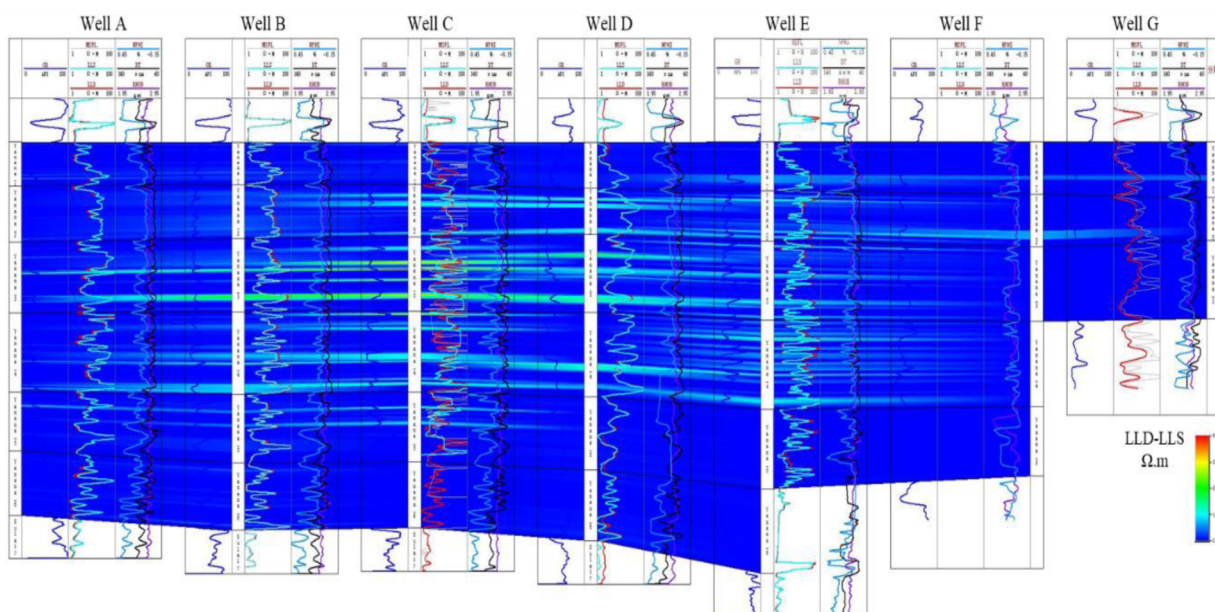


Figure 6—The fracture identification result based on logging interpretation in Yamama reservoir

Fracture prediction and application

Geophysicists pursued to predict the distribution and direction of fractures through seismic data. At present, analysis methods based on seismic attributes, seismic waveforms and seismic velocities have been developed. Among them, seismic attribute analysis is the most widely used prediction method. By calculating curvature attribute, frequency attribute and amplitude attribute, the abnormal characteristics related to fractures are obtained, and then the development of fractures is predicted.

Combination of seismic interpretation and well interpretation is an effective method to predict the spatial distribution of fractures. Based on logging fracture interpretation result, this study optimized the post-stack seismic attributes, including coherence, variance, similarity, symmetry, curvature and ant tracking, etc. The seismic attributes result match well with logging interpretation result are selected as the basis for prediction. By extracting the comprehensive seismic attributes of the target interval, the basic parameters of fractures, such as fracture density, strike and dip are predicted, and the spatial distribution of fractures is clarified.

Fracture interpretation results based on logging interpretation has high accuracy, especially for wells with FMI logging data, so the fractures identified by the well are as the hard data. Comparing the matching relationship between different seismic attributes (coherence, variance, similarity, symmetry, curvature and ant tracking, etc.) and the fracture identification result by the well, the preliminary relationship between fracture development degree and seismic attribute around the wells is determined. Taken the probability volume of fracture development around the well explained by multiple attributes as input data, the fracture development degree and fracture density in the whole area are obtained by CNN method. By comparing with well interpretation fracture result, the input data is continuously optimized to improve the accuracy of fracture prediction.

From the prediction results of fracture density (Figure 7), it shows that except the southwest area, fracture well developed in the study area, especially along the structural axis transition area. Three main fracture groups developed: group1 with the direction of NNW-SSE, located along the wing of the structure; Group2 with the direction of NWW-SEE, along the axis and steeper wing of the main structure; Group3 with the direction of NE-SW, located at the wing of the main structure. The structural axis transition area and the anticlinal core area are well developed fracture, integrating the favorable reservoir prediction result, those area is the favorable appraisal area and suitable for the appraisal well drilling.

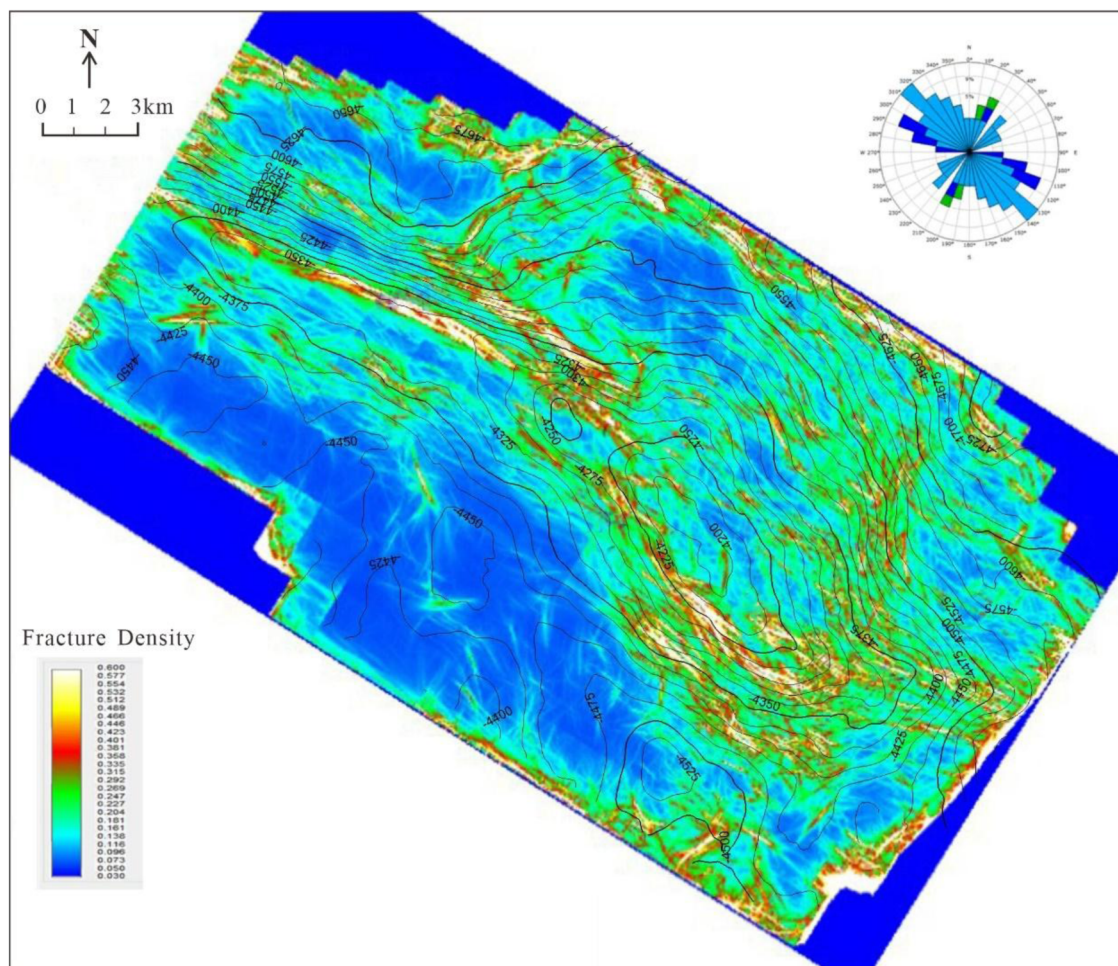


Figure 7—The fracture prediction result (fracture density) in Yamama reservoir

Based on the comprehensive geology analysis and DST test result of drilled wells, the favorable reservoir distribution is determined. Integrating the fracture prediction result, the favorable appraisal area was determined: the favorable area is in the southeast and along the structure axis transition area. According to the predicted results, an appraisal well was deployed in 2023, and good results were achieved in drilling and production. The average production of the appraisal well was up to 4,000 barrels per day. The appraisal well has been stable production for more than two years, with cumulative production up to 3 million barrels.

According to the test results and production performance of wells, the areas with medium fracture density are mostly potential areas, which have good test and production performance. When the fracture density is high, the area may have been severely fractured, similar to the fault zone, the reservoir properties become worse and the distribution is limited. When the fracture density is low, the improvement of reservoir properties is limited and effective reservoirs can't be developed.

Conclusion

1. Yamama reservoir in southeast Iraq has great production potential which have been confirmed by recent exploration and appraisal result in several giant oilfields. Deep burial, low viscosity and abnormal high pressure made it possible to produce more hydrocarbon. Meanwhile, the exploration and appraisal result also revealed that fracture developed in Yamama reservoir. Constrained by regional geological understanding, integrating coring wells, special logging, conventional logging

and seismic data to identify and predict fracture were conducted, the application result confirmed the development and prediction results of fractures.

2. Based on two core wells description, the main lithology of Yamama are limestone and marl, with few shale, the reservoir is tight with poor petrophysical properties. However, the fracture are well developed in Yamama reservoir, the scale is not too large, and most fracture do not cut through the whole core intervals. The fractures are mostly medium-low angle, especially the low-angle or near horizontal fractures are developed. Micro-fractures can be seen in typical thin sections, most of the micro-fractures are open with small scale. The presence of microfractures and small scale fractures can improve the reservoir petrophysical properties.
3. Special logging, especially FMI logging, is an effective method to identify and characterize fractures in wells. The degree of fracture development in each well has big difference. The response characteristics of fractures can also be seen in conventional logging. The big fractures have not developed in Yamama due to non-mud loss during the drilling from masterlog. Based on the calibration of logging response characteristics and core description result, the mud invasion may occur during and after the drill which will obviously show the difference between LLD and LLS logging. The fracture are mainly developed between Well B and Well E, which are located in the structure core and wing area, and the fractures out of the structure core area are not well developed. Vertically, the fracture development interval is located in the middle and upper of Yamama formation. The fracture distributed in a large area, mostly between 2-3 well spacing.
4. Geophysicists pursued to predict the distribution and direction of fractures through seismic data. Combination of seismic interpretation and well interpretation is an effective method to predict the spatial distribution of fractures. Post-stack seismic attributes, including coherence, variance, similarity, symmetry, curvature and ant tracking, etc are used to predict fracture. The predict result shows that except the southwest area, fracture well developed in the study area, especially along the structural axis transition area. Three main fracture groups developed: group1 with the direction of NNW-SSE, located along the wing of the structure; Group2 with the direction of NWW-SEE, along the axis and steeper wing of the main structure; Group3 with the direction of NE-SW, located at the wing of the main structure. The structural axis transition area and the anticlinal core area are well developed fracture.

This study conducted comprehensive research on fracture of Yamama reservoir in southeast Iraq. It aims to discuss the identification and characterization of Yamama reservoir fractures, which will provide guidance for efficient exploration and development of this reservoir in the future.

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